

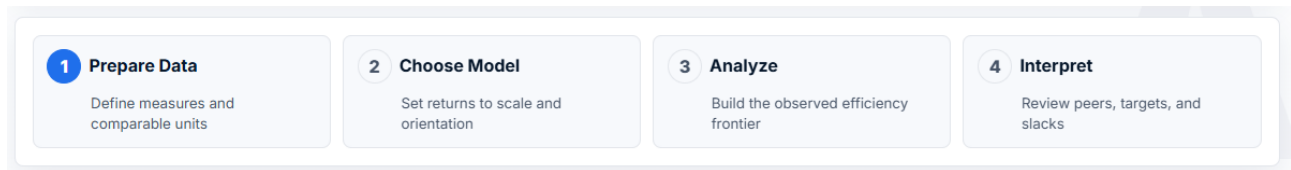
PRACTICAL GUIDE

From Warehouse KPIs to Efficiency-Based Planning

How to use Data Envelopment Analysis (DEA) in Analytical Tools Hub

A guide to build a historical efficiency frontier, interpret peers and scale effects, then use the fixed frontier to support future input or output benchmarking.

Traditional KPIs tell us what happened. DEA helps us ask how efficiently multiple resources were transformed into multiple operational outcomes.



Warehouse illustration: 12 monthly DMUs | Inputs: Labour Hours, Overtime Hours | Outputs: Orders Shipped, Lines Picked

**This data is generated by Gen AI for illustration purposes*

Month	Labour Hours	Overtime Hours	Orders Shipped	Lines Picked
Apr	1610	170	31800	74500
May	1570	125	32400	76800
Jun	1720	220	34900	80200
Jul	1650	155	34300	81400
Aug	1790	245	36700	84600
Sep	1700	180	35800	85300
Oct	1880	290	38200	88100
Nov	2080	450	42600	97500
Dec	2260	610	45800	104200
Jan	1850	275	37600	87300
Feb	1690	160	35200	82900
Mar	1760	205	37100	86800

1. Prepare comparable historical data

DEA is most useful when the decision-making units (DMUs) perform the same broad function and are measured with consistent definitions. For this warehouse example, each month is one DMU.

Blank custom model

Supply chain and logistics

- Distribution Centre Efficiency
- Warehouse Operations Efficiency
- Supplier Delivery Efficiency
- Transport Depot Efficiency
- Last-Mile Delivery Hub Efficiency
- Procurement Team Efficiency

Manufacturing and asset operations

- Manufacturing Plant Efficiency
- Production Line Efficiency
- Maintenance Team Efficiency
- Quality-Control Unit Efficiency

Commercial and service operations

- Retail Branch Efficiency
- Regional Sales Office Efficiency
- Customer Support Centre Efficiency
- Service Centre Efficiency

Blank custom model ▼

Load Sample Template **Reset**

Select an illustrative operating scenario, then review and adapt its measures, units, assumptions, and data before analysis.

Sample Output Interpretation

A score below 100% indicates a unit is below peers and targets show what comparable

3 Analyze

Build the observed efficiency frontier

with consistent definitions.

Start from a sample template or a blank custom model; templates are illustrative and should be adapted.

Define the measures

INPUTS	OUTPUTS
Labour Hours	Orders Shipped
Overtime Hours	Lines Picked

Input measures **Add Input**

Labour Hours X

Overtime Hours X

Output measures **Add Output**

Orders Shipped X

Lines Picked X

Two resource inputs and two throughput outputs keep the demonstration easy to interpret.

Data quality matters

- Use one row per DMU and consistent units/definitions.
- Check missing, negative, zero or invalid values.
- Avoid too many measures for too few DMUs; weak discrimination can make too many units appear efficient.

2. Enter or import the warehouse DMUs

Enter values manually or import a CSV. Configure the correct number of input and output measures before import, then review the values before analysis.

Decision-making units

At least two units are required. More observations generally create a more informative comparison.

Add
DMU

DMU NAME	LABOUR HOURS INPUT	OVERTIME HOURS INPUT	ORDERS SHIPPED OUTPUT	LINES PICKED OUTPUT	REMOVE
Apr	1610	170	31800	74500	X
May	1570	125	32400	76800	X
Jun	1720	220	34900	80200	X
Jul	1650	155	34300	81400	X
Aug	1790	245	36700	84600	X
Sep	1700	180	35800	85300	X
Oct	1880	290	38200	88100	X
Nov	2080	450	42600	97500	X
Dec	2260	610	45800	104200	X

Illustrative monthly warehouse DMUs.

Comparability comes before calculation. A DEA score is only meaningful when the periods or units being compared are operationally comparable.

For this illustration

- 12 historical months form the reference dataset.
- Labour Hours and Overtime Hours represent resource consumption.
- Orders Shipped and Lines Picked represent throughput.
- A real implementation may also need quality, service and workload-complexity measures.

3. Choose the DEA model

The model choice should follow the management question.

Returns to scale

BCC / VRS - variable returns to scale

BCC separates scale effects when units operate at different sizes. CCR assumes outputs can change proportionally with inputs.

Orientation

Input-oriented - reduce resources

Input orientation asks how much resource use could fall. Output orientation asks how much output could rise with current resources.

Selected approach:

BCC separates scale effects by allowing variable returns to scale and focuses on proportional input reduction while maintaining current outputs.

Warehouse example: BCC / VRS with an input-oriented model.

Returns to scale

Model	Assumption	Use when
CCR / CRS	Constant returns to scale	Scale effects are not being separated.
BCC / VRS	Variable returns to scale	DMUs may operate at different sizes; separate scale effects.

Orientation

- Input-oriented: Can the same outputs be produced with fewer inputs?
- Output-oriented: Can more outputs be produced with the same inputs?

For the warehouse case, BCC / VRS + input orientation fits a resource-planning question: Could the observed outputs have been achieved with fewer Labour Hours and Overtime Hours?

4. Run DEA and interpret the historical frontier

3. Run DEA

Validate the prepared data and build the observed efficiency frontier using the selected model.

Run DEA

Run DEA after validating the data and assumptions.

The tool evaluates Labour Hours and Overtime Hours simultaneously against Orders Shipped and Lines Picked. This differs from choosing a single ratio such as orders per labour hour.

4. Results and Interpretation

Efficiency is relative to the best observed combinations in this dataset, not an absolute performance standard.

[Export Results CSV](#)

[Print Report](#)

DMUS COMPARED

12

EFFICIENT UNITS

6

AVERAGE EFFICIENCY

98.7%

LOWEST EFFICIENCY

96.7%

Headline results summarize the number of DMUs, frontier units and relative efficiency.

EXECUTIVE BENCHMARK SUMMARY

6 units need benchmark review

Jan has the lowest score at 96.7%. The model is providing useful discrimination across the sample.

BEST-PRACTICE SET

frontier units

6

MOST USED PEER

3 references

May

MAIN SLACK SIGNAL

output shortfall in Jun

Lines Picked

The executive benchmark summary highlights peer and slack signals.

A 100% score means efficient relative to the selected sample and model.

It does not mean perfect performance in an absolute sense.

5. Go beyond the score: peers, targets and slacks

The ranking is the starting point. For an inefficient DMU, inspect which efficient units form its benchmark, what target changes are implied and whether non-radial slacks remain.

Efficiency Ranking				
RANK	DMU	EFFICIENCY	STATUS	BENCHMARK PEERS
1	May	100.0%	Efficient frontier	May
1	Sep	100.0%	Efficient frontier	Sep
1	Nov	100.0%	Efficient frontier	Nov
1	Dec	100.0%	Efficient frontier	Dec
1	Feb	100.0%	Efficient frontier	Feb
1	Mar	100.0%	Efficient frontier	Mar
7	Jul	99.7%	Improvement opportunity	May, Sep, Feb
8	Apr	97.5%	Improvement opportunity	May
9	Aug	97.3%	Improvement opportunity	Sep, Mar
10	Oct	97.0%	Improvement opportunity	Dec, Mar
11	Jun	96.8%	Improvement opportunity	May, Sep
12	Jan	96.7%	Improvement opportunity	Dec, Mar

Efficiency ranking identifies frontier units and improvement opportunities.

Peer Benchmark and Improvement Targets

Select a unit to inspect its benchmark peers, targets, and non-radial slacks.

Decision-making unit

Jul - 99.7%

Jul: The radial score suggests up to 0.3% proportional input reduction before considering slacks. Peer reference: May (0.419), Sep (0.457), Feb (0.124)

PEER MIX

May

λ 0.4193

Sep

λ 0.4568

Feb

λ 0.1239

BENCHMARK SIGNAL

Lines Picked

Largest output target gap: 38.49.

MANAGEMENT READING

Review whether inputs could be reduced by about 0.3% while maintaining current outputs. Investigate the additional Lines Picked shortfall beyond the proportional adjustment. Compare operating practices with May, Sep, Feb before setting a management target.

TYPE	MEASURE	ACTUAL	PEER-BASED TARGET	CHANGE	SLACK
Input	Labour Hours	1,650	1,644.25	-0.3%	0
Input	Overtime Hours	155	154.46	-0.3%	0
Output	Orders Shipped	34,300	34,300	-0.0%	0
Output	Lines Picked	81,400	81,438.49	+0.0%	38.49

Peer Benchmark and Improvement Targets show peer weights, targets and slacks.

- Which efficient months are repeatedly used as peers?
- What operating conditions made those periods different?
- Are the target reductions operationally realistic?

- Do slacks reveal a specific resource or output needing separate attention?

6. Understand scale efficiency and model sensitivity

Technical efficiency alone does not answer whether the warehouse is operating at an appropriate scale. Comparing CCR and BCC helps separate overall technical efficiency from scale effects.

CCR versus BCC				
Uses the baseline input-orientation and changes only the returns-to-scale assumption.				
DMU	CCR EFFICIENCY	BCC EFFICIENCY	SCALE EFFICIENCY	INTERPRETATION SIGNAL
Apr	93.8%	97.5%	96.2%	Review operating scale
May	100.0%	100.0%	100.0%	Limited scale inefficiency
Jun	96.3%	96.8%	99.4%	Limited scale inefficiency
Jul	99.6%	99.7%	100.0%	Limited scale inefficiency
Aug	97.3%	97.3%	100.0%	Limited scale inefficiency
Sep	100.0%	100.0%	100.0%	Limited scale inefficiency
Oct	96.4%	97.0%	99.4%	Limited scale inefficiency
Nov	97.2%	100.0%	97.2%	Review operating scale
Dec	96.1%	100.0%	96.1%	Review operating scale
Jan	96.4%	96.7%	99.7%	Limited scale inefficiency
Feb	99.8%	100.0%	99.8%	Limited scale inefficiency
Mar	100.0%	100.0%	100.0%	Limited scale inefficiency

CCR versus BCC comparison provides a scale-efficiency signal.

Assumption Sensitivity

3 of 12 DMUs have scale efficiency below 98%. Review whether operating size contributes to the gap for Apr, Nov, Dec.

1 of 12 DMUs change by more than 2 percentage points between input and output orientation. Clarify whether management can primarily control resources or outcomes for Apr.

Scenario comparisons test modelling assumptions only. They do not prove which specification is correct, and they do not overwrite the baseline ranking, peers, targets, or export above.

Assumption sensitivity flags where scale or orientation changes the interpretation.

Sensitivity analysis does not prove one specification is 'correct'. It shows whether the management conclusion depends heavily on the modelling assumption.

- **Technical Efficiency:** how effectively resources are converted into outputs.
- **Scale Efficiency:** whether part of the gap is associated with operating scale.
- **Returns to Scale:** whether output changes proportionally, more than proportionally, or less than proportionally as scale changes.

7. Move from historical analysis to future planning

Scenario Analysis uses the historical DMUs as the fixed reference set. The future scenario is evaluated against that frontier; it does not construct or alter it.

☒ **Generate Input Requirement**
Enter target or forecast outputs. DEA generates the efficient benchmark inputs implied by the fixed frontier.

☐ **Generate Output Requirement**
Enter available inputs. DEA generates the efficient benchmark outputs implied by the fixed frontier.

Choose the direction of the planning question.

Planning question	You provide	DEA generates
Generate Input Requirement	Target / forecast outputs	Efficient benchmark inputs
Generate Output Requirement	Available inputs	Efficient benchmark outputs

Outputs known → benchmark the inputs. | Inputs known → benchmark the outputs.

8. Warehouse planning scenario

Suppose next month's forecast is:

40,000	92,000
Orders Shipped	Lines Picked

☒ **Generate Input Requirement**
Enter target or forecast outputs. DEA generates the efficient benchmark inputs implied by the fixed frontier.

☐ **Generate Output Requirement**
Enter available inputs. DEA generates the efficient benchmark outputs implied by the fixed frontier.

Scenario Values SCENARIO

Enter target or forecast outputs using the same definitions as the historical data.

Scenario name

Future Scenario Plan

Orders Shipped
Target / forecast output

Lines Picked
Target / forecast output

40000

92003

Evaluation assumptions

Choose whether the fixed frontier should use constant or variable returns to scale.

Returns to scale

BCC - variable returns

Generate Benchmark Inputs

Enter forecast outputs, select the returns-to-scale assumption, then generate benchmark inputs.

The planning question changes

Instead of: “Demand is increasing by X%, so let's increase labour by X%.”

Ask: “Given the outputs we need to achieve, what input requirement is supported by our historically efficient operating performance?”

The values generated are a DEA Efficient Benchmark, not a guaranteed staffing forecast.

Planners should still consider shift patterns, skill mix, SKU complexity, automation, service levels, equipment constraints and demand variability.

9. What DEA adds to the management conversation

DEA does not replace warehouse KPIs. It adds a multi-input, multi-output benchmarking lens that can help managers challenge resource assumptions, identify efficient peers and examine whether operating scale contributes to the observed gap.

Before scaling resources with demand, first ask whether the historical performance you are scaling is actually efficient.

Use the tool to support four conversations

- **Historical efficiency:** Which periods or units form the observed frontier?
- **Improvement:** Which peers, targets and slacks deserve investigation?
- **Scale:** Is inefficiency associated with resource conversion, operating scale, or both?
- **Planning:** Given forecast outputs or available inputs, what benchmark does the fixed historical frontier imply?

Important interpretation note:

DEA results are relative to the selected sample, measures and model assumptions. Combine them with operational context rather than using them as an automatic staffing or performance decision.

**Explore the Data Envelopment Analysis Tool in
Analytical Tools Hub**

[analyticaltoolshub.com/Data Envelopment Analysis/Data_Envelopment_Analysis.html](https://analyticaltoolshub.com/Data%20Envelopment%20Analysis/Data_Envelopment_Analysis.html)